

Hema Latha

mh1atha22@gmail.com | +1 (704) 307-8048 | linkedin.com/in/hema-mannam | https://github.com/Hemamannam

SUMMARY

Software Engineer with 1+ years of hands on experience across **industry internships, production systems, and real world projects**. Strong foundation in **object oriented programming (OOP), data structures, algorithms and distributed systems**, with practical experience building **scalable, low latency backend and full stack applications**. Proficient in **Java, Python and JavaScript**, with experience developing, testing and deploying production ready services on cloud platforms with a focus on **performance, reliability and observability**.

PROFESSIONAL EXPERIENCE

Software Engineer Intern – Backend, Linxy Live

07/2025 – Present | Remote, USA

- Built and optimized **performance critical, real time backend systems**, applying **performance profiling, latency analysis, and reliability focused optimizations** to **reduce end to end latency by 35%**.
- Designed and evaluated **machine learning** based algorithms using **structured experimentation**, translating functional requirements into deployable production logic and **improving system effectiveness and user engagement by 40%+**.
- Architected modular **backend services, RESTful APIs and distributed microservices**, emphasizing **low latency execution, fault tolerance and high availability**, achieving **99.9% service uptime** under peak load.
- Implemented **AI driven notification and alerting systems** to intelligently trigger real time notifications based on system events and user behavior, improving incident response time and operational awareness.
- Deployed and operated **cloud native backend services** on **AWS, leveraging CI/CD pipelines** to ensure **reliable releases**, improved **system stability and consistent deployment workflows**.

Software Engineer Intern, Nalsoft

08/2023 – 07/2024 | Hyderabad, India

- Built and maintained **Java based backend services** for enterprise **applications, focusing on reliability and performance**.
- Implemented **REST APIs** and optimized **database queries using SQL** to reduce end to end request **latency by 35%**.
- Worked with **relational databases** to support **transactional workflows** and **reporting features in enterprise applications**.
- Debugged **backend services and production issues**, contributing **fixes and performance improvements** in a distributed environment.
- Collaborated with cross functional teams to **implement automated testing and CI/CD pipelines**, ensuring **high quality releases**.

TECHNICAL SKILLS

Programming Languages: Java, Python, C#, C++, Go, JavaScript, TypeScript, SQL, Spring Boot, .NET, React.js, Angular, HTML5, CSS3, Sass

Quantitative & Statistical: Probability, Statistical Analysis, Regression, Time Series Analysis (TSA), Artificial Neural Networks , RNN, LSTM

Machine Learning & Data Visualization: Scikit Learn, TensorFlow, Computer Vision, NLP, Matplotlib, Seaborn, Power BI, Tableau

Systems & Networking: Low Latency Systems, High Performance Computing, Multithreading, Lock-Free Data Structures, TCP/ IP, UDP

Databases: Relational Databases (MySQL, PostgreSQL), NoSQL Databases (MongoDB, DynamoDB)

Technical Domains & Tools: Data Analytics, Deep Learning, Generative AI (LLM concepts & experimentation), Cloud Computing (AWS EC2, S3; GCP; Azure), Agile/Scrum, Extreme Programming (XP), Unit Testing (JUnit, pytest), CI/CD, Docker, Git, GitHub, Linux, Jira

PROJECTS

Movie Night | Java, Spring Boot, MySQL, JavaScript

- Delivered a user facing **full stack web application** with role based authentication and **RESTful backend services** using **Spring Boot** and **MySQL**, optimizing **API performance and database queries by 40%**.
- Enabled real time data ingestion and extensible, API first system capabilities by integrating **third party APIs**, supporting analytics driven features and improving user productivity.
- Accelerated **feature delivery by 30%** through **Agile development**, Git based collaboration, and automated testing, resulting in higher code quality, reliability, and long term maintainability.

3D Facial Expression Reconstruction | Python, PyTorch, OpenCV

- Engineered and executed **deep learning experiments** for **3D facial reconstruction** using **SPECTRE and DECA**, applying linear algebra based representations and probabilistic loss functions.
- Performed **quantitative and qualitative evaluations** on identity **preservation, achieving 95% accuracy** and analyzing failure cases across diverse expressions and viewpoints.
- Optimized and iteratively refined **GPU accelerated inference pipelines**, experimenting with architectural and hyper parameter variations to improve latency and reconstruction quality

SignSpeak – Hand Gesture Recognition System | Python, OpenCV, MediaPipe, TensorFlow, NLP, TTS

- Developed a **real-time sign language recognition system** using **Python, MediaPipe, OpenCV, and TensorFlow**, supporting **30+ gestures** with live webcam input.
- Trained and deployed **CNN/RNN models** achieving **90%+ accuracy** and **<200ms inference latency** for real time gesture classification.
- Integrated **NLP and Text to Speech pipelines** to enable **gesture to text and gesture to audio conversion**, enhancing accessibility.

EDUCATION

Master of Science in Computer Science, University of North Carolina at Charlotte

AUG 2024 – DEC 2025

CGPA: 3.8/4.0

Bachelor of Technology in Computer Science, SRM University | CGPA: 3.6 / 4.0

CERTIFICATES

- AWS Academy – Cloud Foundations
- AWS Academy – Data Center Technician
- Microsoft Azure Fundamentals (AZ-900)
- Generative AI with Large Language Models